

HOCHSCHULE HAMM-LIPPSTADT (HSHL)

Hardware Equipment Recommendation Report

Band01_Rev1 Smartband · Task 3 Preparation

Submitted to : Prof. Dr. Javidi da Costa, Joao Paulo

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1. Purpose of This Report

This report provides a formal hardware procurement recommendation for the Band01_Rev1 smartband project at Hochschule Hamm-Lippstadt (HSHL). It is intended to give the supervising professor a clear, technically justified list of items to order so that the project can progress from its current state — firmware compiled and ready — to a state where the firmware can be flashed onto the physical PCB and sensor readings can be tested and verified.

The recommendations are based on thorough analysis of the following sources:

- The Band01_Rev1 PCB schematic (Altium Designer, Ricardo K. & Gabriel P., 2019)
- The Aisler Bill of Materials for Band01_Rev1 (March 2026)
- Physical photographs of the two manufactured PCB panels delivered to HSHL
- The firmware repository (smartband_v001_firmware_masrur) and its build system
- Practical build testing performed on 28 May 2026 — firmware successfully compiled

Current project status: PCB manufactured and physically available · Firmware compiled successfully on 28 May 2026 · Two missing repository files identified and fixed (build_config.opt, gapbondmgr.h) · No debug probe available · No power supply defined for bench testing

2. Critical Power Architecture Analysis

Before any power supply can be recommended, the Band01_Rev1 power architecture must be understood from the schematic. An incorrect power connection will permanently damage the LP5907 regulator on the PCB. This section is the technical foundation for all power recommendations in Section 4.

2.1 How the Board is Powered

The Band01_Rev1 uses a dual-rail power architecture. A single-cell LiPo battery connects to the VBAT net, which feeds two LP5907 ultra-low-noise LDO regulators. U101 produces a clean 3.3V rail (NL3V3) powering the CC2640R2F, LIS331 accelerometer, MAX30205 temperature

sensor and I²C pull-up resistors. U102 produces a separate 1.8V rail (NL1V8) powering the MAX30102 heart-rate sensor's analogue front end exclusively.

P1 Pin	Net	What it connects to — confirmed from Altium schematic
Pin 12	VBAT	Direct input to LP5907 U101 and U102. This is the battery positive terminal — the correct pin to power the board externally at 3.3V–3.8V.
Pin 18	3V3	NL3V3 net — the OUTPUT of LP5907 U101. Shared with CC2640R2F, LIS331, MAX30205, I ² C pull-ups. Do NOT connect power to this pin.
Pin 17	GND	Ground — connect negative of any power source here.
Pin 16	N/C	Not connected in schematic — do not use.

Why AMS1117-3.3 will damage this PCB: An AMS1117 module outputs regulated 3.3V. Connecting it to P1 pin 18 (3V3 net) injects current into the OUTPUT pin of LP5907 U101. The LP5907 datasheet does not permit reverse current on its output — this will damage or destroy U101. The correct pin is P1 pin 12 (VBAT), which enters the circuit before the regulators exactly as a battery would.

3. Debug Probe Recommendations

A debug probe connects between the development laptop and the Band01_Rev1 P1 header via JTAG. It allows Code Composer Studio to flash firmware to the CC2640R2F, set breakpoints, and inspect memory in real time. The university's original XDS110 probe is confirmed out of stock. Three alternatives are provided in ranked order.

JTAG wiring — identical for all three options: Probe VTref → P1 pin 18 (3V3, sense only — does not supply power) · GND → P1 pin 17 · TDI → P1 pin 11 · TMS → P1 pin 9 · TCK → P1 pin 7 · TDO → P1 pin 13 · nRESET → P1 pin 15

1 SEGGER J-Link EDU Mini — RECOMMENDED

Property	Detail
Part number	8.08.91 (SEGGER Microcontroller GmbH)
Interface	JTAG and SWD — both supported. CC2640R2F uses JTAG.
CCS configuration	Step 1: Install SEGGER J-Link Software from segger.com (installs USB drivers). Step 2: In CCS open targetConfigs/CC2640R2F.ccxml → change Connection from 'XDS110' to 'SEGGER J-Link Emulator' → Save.
License	EDU license — explicitly granted for educational and non-commercial university use. This HSHL student project qualifies without restriction.

Physical	Small flat card form factor with 10-pin ribbon cable. Compact, portable, fits easily in a lab bag.
Why it is the best choice	Compact, widely available in Germany with fast delivery. Fully proven with CC2640R2F. Natively supported by CCS. Correct license for academic use. The SEGGER J-Link software also includes a standalone commander tool for flashing without CCS — useful for quick hardware verification without opening the full IDE.



Figure 1 — SEGGER J-Link EDU Mini (Part: 8.08.91) with 10-pin ribbon cable and JTAG connector

 [Buy on Mouser Germany — mouser.de](https://www.mouser.de) (Part: 829-8.08.91)

2 LAUNCHXL-CC2640R2 — ALTERNATIVE (if Option 1 unavailable)

Property	Detail
Product	LAUNCHXL-CC2640R2 — CC2640R2F LaunchPad development kit (Texas Instruments)
How to use as probe	Remove the isolation jumpers on the LaunchPad to disconnect its onboard CC2640R2F. Connect the JTAG header to Band01_Rev1 P1 using the wiring table above. CCS detects it as a standard XDS110 probe.
CCS configuration	No changes needed — the project was designed for XDS110. The existing targetConfigs/CC2640R2F.ccxml connection works out of the box with no edits.
What it lacks vs Option 1	Significantly larger and bulkier. You are buying a full development board but only using its onboard XDS110 debugger chip. The LaunchPad's own CC2640R2F is unused and wasted. More expensive than the J-Link Mini. Requires removing isolation jumpers before use as a probe.
When to prefer this	If J-Link EDU Mini is unavailable everywhere. Also useful if the project later needs a reference CC2640R2F platform to compare firmware behaviour against Band01_Rev1.



Figure 2 — LAUNCHXL-CC2640R2 LaunchPad development kit with USB cable (Texas Instruments)

Buy on Mouser Germany — [mouser.de](https://www.mouser.de) (LAUNCHXL-CC2640R2)

3 SEGGER J-Link BASE — ALTERNATIVE (commercial license version)

Property	Detail
Part number	8.08.28 (SEGGER Microcontroller GmbH)
Difference from Option 1	Electrically and functionally identical to the EDU Mini for this project. The only difference is licensing — BASE carries a full commercial license permitting use in commercial and industrial projects without restriction.
CCS configuration	Identical to Option 1 — install SEGGER J-Link software, change Connection to 'SEGGER J-Link Emulator' in targetConfigs/CC2640R2F.ccxml.
Physical	Larger black rectangular form factor compared to the flat EDU Mini card. Includes a 20-pin JTAG connector with ribbon cable.
What it lacks vs Option 1	Significantly more expensive. The commercial license is completely unnecessary for an HSHL academic student project — the EDU license of Option 1 already covers all legitimate uses here. No technical advantage for this use case.
When to prefer this	Only if HSHL lab policy explicitly requires commercially licensed equipment for all projects regardless of type, or if the probe will be shared across commercial work beyond this project.



Figure 3 — SEGGER J-Link BASE / Plus (Part: 8.08.28) — commercial license version

 Buy on Mouser Germany — [mouser.de](https://www.mouser.de) (Part: 829-8.08.28)

4. Power Supply Recommendations

The board must be powered via P1 pin 12 (VBAT) as established in Section 2. Two safe options are provided. Option 1 is the correct long-term solution and is required for the final wearable demonstration. Option 2 is suited for stable bench testing before a battery is permanently connected.

1 Single-Cell LiPo Battery 3.7V 150mAh — RECOMMENDED

Property	Detail
Specification	Single-cell lithium polymer (LiPo), 3.7V nominal, 150mAh capacity
Connector	Bare wire leads (red = positive, black = negative) — visible in product image. Verify with Giovanni whether a JST-PH 2.0mm connector footprint is soldered on the PCB. If not, solder wires directly to VBAT and GND pads on the PCB bottom side.
Connection	Red wire (positive) → P1 pin 12 (VBAT) Black wire (negative) → P1 pin 17 (GND)
Why it is the best choice	This is exactly how the board was designed to operate — VBAT feeds the LP5907 regulators as intended by the schematic. Delivers 73+ hours of continuous operation at 2.032 mA average current (Pineiro et al. 2021). Required anyway for the final wearable demonstration where the board must be untethered from a desk.
What it lacks	Requires a separate LiPo charger module between sessions (e.g. TP4056 module, ~€1). Battery voltage drops slightly over long test sessions — bench supply maintains more stable voltage during extended development work.



Figure 4 — Single-cell LiPo battery 3.7V 150mAh with bare wire leads (red = +, black = -)

 [Buy on Reichelt Elektronik — reichelt.de \(LiPo 150mAh 3.7V\)](https://www.reichelt.de/LiPo-150mAh-3.7V)

2 Adjustable Bench Power Supply set to 3.7V → P1 pin 12 — ALTERNATIVE

Property	Detail
Specification	Adjustable bench power supply capable of 3.3V–3.8V output with at least 200mA current. The unit shown includes a colour display showing voltage, current and power, with built-in test leads.
Connection	Red lead → P1 pin 12 (VBAT) Black lead → P1 pin 17 (GND) Set voltage to 3.7V and current limit to 150mA before connecting
Why it works	Simulates the LiPo battery at VBAT — enters the circuit before the LP5907 regulators exactly as the battery would. The LP5907 regulators produce the correct 3.3V and 1.8V rails internally. Provides stable, controlled voltage throughout a test session with no drop.
What it lacks vs Option 1	The board cannot be worn or moved during testing — it must remain on the desk tethered to the supply. Not suitable for the final wearable demonstration. If the university supply minimum is 5V, an additional step-down buck converter (LM2596 or MP1584 adjustable module) is needed between supply output and VBAT.
When to prefer this	For initial firmware flashing and first sensor activation sessions where stable controlled voltage and easy power cycling are more important than portability.



Figure 5 — Adjustable bench power supply with colour display, 5V input, adjustable output 0–30V with test leads

 [Search on Amazon Germany — amazon.de](https://www.amazon.de)

Can both power sources be connected simultaneously? NO — never connect both a LiPo battery and a bench supply to VBAT at the same time. Two voltage sources on the same net will conflict and may damage the battery, the LP5907 regulators, or both. Use exactly one power source at a time.

5. Required Wiring & Accessories

The following items are required regardless of which probe or power option is chosen.

Item	Purpose	Supplier
Dupont jumper wires M-F 20cm (pack of 40)	Connect debug probe to P1 header (7 wires for full JTAG) · Connect power supply to P1 pin 12 (VBAT) and P1 pin 17 (GND)	Amazon.de
USB-A cable	Connect J-Link EDU Mini or LaunchPad to laptop USB port	Any supplier
TP4056 LiPo charger module (optional)	Recharge the LiPo battery between test sessions. Input: USB 5V. Output: charges single-cell LiPo safely.	Amazon.de

6. Complete Build and Flash Procedure

Steps 1–5 are the one-time build environment setup (already completed on this machine on 28 May 2026). Steps 6–10 require the physical hardware and must be performed once the ordered items arrive.

Build Environment (already completed — for reference):

1. Clone repository to C:\ti\ and open CCS. Switch to workspace_v12.
2. TOOLS/build_config.opt and Include/gapbondmgr.h are now fixed in the repo — no manual creation needed.
3. In workspace_v13: import and build simple_peripheral_cc2640r2lp_stack_library from C:\ti\simplelink_cc2640r2_sdk_5_30_01_11\examples\rtos\CC2640R2_LAUNCHXL\blestack\simple_peripheral\tirtos\ccs\
4. Copy the three output files (.lib, lib_linker.cmd, ble_r2.symbols) from the stack library's FlashROM_Library/ folder to C:\FlashROM_Library\
5. Build smart_band_v001_app in workspace_v12. Should compile with 0 errors.

Hardware Setup (requires ordered items):

6. If using J-Link (Options 1 or 3): Install SEGGER J-Link Software from segger.com. In CCS open targetConfigs/CC2640R2F.ccxml → change Connection to 'SEGGER J-Link Emulator' → Save.
7. If using LaunchPad (Option 2): Remove isolation jumpers. No CCS changes needed.
8. Set power supply to 3.7V (or connect LiPo). Connect positive to P1 pin 12 (VBAT) and negative to P1 pin 17 (GND). Connect probe to P1 header. Connect probe USB to laptop.
9. In CCS: Run → Debug. CCS flashes firmware and halts at main(). Press Play to run.
10. Verify on smartphone with nRF Connect: connect to 'Simple BLE Peripheral' and confirm CHAR1 (UUID 0xFFF1) is transmitting. Currently sends placeholder {0,1,2,0,0x77,0x88} — live sensor activation is the next step.

7. Summary — Items to Order

All items required to flash firmware, activate sensors, and run the final wearable demonstration:

#	Item	Recommended Product	Where to Buy (Germany)	Priority	Price
1	Debug Probe	SEGGER J-Link EDU Mini (8.08.91)	https://www.antratek.de/j-link-edu-mini	Essential	68.96€
2	Power Supply	3.7V 150mAh LiPo with bare leads	https://shorturl.at/6IAQM	Essential	4.99€
3	Jumper Wires	Dupont M-F 20cm (pack of 40)	https://shorturl.at/QKwP6	Essential	8.49€
4	USB Cable	Standard USB-A cable	https://shorturl.at/A7Gu5	Essential	1.20€
5	LiPo Charger	TP4056 single-cell charger module	https://shorturl.at/9NoGK	Recommended	6.68€
6	Debug Probe	LAUNCHXL-CC2640R2	https://shorturl.at/MxydN	Alternative	32.42€
7	Debug Probe	Segger J Link Base	https://rb.gy/2wpfeo	Alternative	589.05
8	Power Supply	FNIRSI® DPS-150	https://h1.nu/1vTDc	Alternative	61.95€

Raspberry Pi note: No additional hardware purchase is required for Raspberry Pi integration. Once firmware is running, the Pi connects wirelessly via BLE using Python (bluepy or bleak library) to receive GATT notifications from CHAR1 (UUID 0xFFFF1) every 350ms. No physical wiring is needed for this integration.

8. References

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